

Yasod Ginige

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SUMMARY AND RESEARCH INTERESTS

A highly motivated individual who is passionate about research that addresses real-world challenges and creates meaningful impact. Currently pursuing a PhD in Computer Science at the University of Sydney, with research interests in **Machine Learning**, **Computer Networks and Security**, and **Computer Vision**. I am committed to continuously learning new concepts and developing innovative solutions to complex problems.

EDUCATION

- Jan. 2024 - Present **PhD in Computer Science** - University of Sydney, Australia
- Title : AI Enhanced Cybersecurity for Red Teaming and Blue Teaming Tasks.
 - Build autonomous cybersecurity tools integrating AI technologies with security tools. The aim is to cater the skill shortage in security industry and increase the usability of security tools for the general public.
 - Finalist in the **Google Student Fellowship**. Was one of the two nominees of the University of Sydney.
 - Received funding from the USYD Collaborative Research Equipment Grant and the NHMRC Equipment Grant Schemes.
- Nov. 2018 - Jul 2023 **B.Sc.(Hons) in Electronic and Telecommunication Engineering** - University of Moratuwa, Sri Lanka
- Cumulative GPA : 4.05/4.20 (*First Class Honors*) - in **top 10 performers**.
 - Was nominated for the Most Outstanding Graduate of the Year.
 - Cricket team member, Mathematics club president, and IEEE IES chapter president.
- Jul. 2015 - Aug. 2017 **RICHMOND COLLEGE - GALLE, Sri Lanka**
- The most outstanding student of the year 2017.
 - **National Rank : 8th** in the G.C.E. Advanced Level Examination (from ~20,000 candidates).
 - Honorable mention award in the International Physics Olympiad, 2018.
 - Gold medal in the Sri Lankan Physics Olympiad, 2017.

PUBLICATIONS

- **Pen-Strategist : A Reasoning Framework for Penetration Testing Strategy Formation and Analysis** [pdf]
Ginige Y., Marasinghe P., Jain S., Seneviratne S. *Under Review*
- **SADE : Symptom-Aware Diagnostic Escalation for LLM-Based Network Troubleshooting** [pdf]
Tseng K., Bogahawatta N., Ginige Y., Dekic K., Sivanathan A., Seneviratne S. *Under Review*
- **AutoPentester : An LLM Agent-Based Framework for Automated Pentesting** [pdf]
Ginige Y., Akila N., Jain S., Seneviratne S. *IEEE TrustCom 2025*
- **TrafficLLM : LLMs for Improved Open-Set Encrypted Traffic Analysis** [pdf]
Ginige Y., Silva B., Dahanayaka T., Seneviratne S. *Computer Networks, 2025*
- **Robust Open-Set Classification for Encrypted Traffic Fingerprinting** [pdf]
Dahanayaka T., Ginige Y., Huang Y., Jourjon G., Seneviratne S. *Computer Networks, 2023*
- **Quantifying and Exploiting Adversarial Vulnerability : Gradient-Based Input Pre-Filtering for Enhanced Performance in Black-Box Attacks** [pdf]
Karunanayake N., Silva B., Ginige Y., Seneviratne S., Chawla S. *ACM TOPS, 2025*
- **TrafficGPT : An LLM Approach for Open-Set Encrypted Traffic Classification** [pdf]
Ginige Y., Dahanayaka T., Seneviratne S. *Asian Internet Engineering Conference, 2024*
- **VC-FeS : Viewpoint-Conditioned Feature Selection for Vehicle Re-identification in Thermal Vision** [pdf]
Ginige Y., Gunasekara R., Hevawithara D., Ariyaratne M., Rodrigo R., Jayasekara P. *Machine Learning and Cybernetics, 2026*
- **A Framework to Assess Multilingual Vulnerabilities of LLMs** [pdf]
Tang L., Bogahawatta N., Ginige Y., Xu J., Sun S., Ranathunga S., Seneviratne S. *WWW Conference, 2025*

ARTICLES

- E-Carrier magazine : Official magazine of the Department of Electronic and Telecommunication Engineering
Published two articles, namely "[Silicon to DNA](#)" and "[Website Traffic Fingerprinting](#)".

RESEARCH EXPERIENCE

PHD RESEARCH

University of Sydney, Australia

LLM Agents Based Autonomous Penetration Testing [\[code\]](#)

2024 - 2025

- Developed an automated pentesting tool based on LLMs and ML techniques to address the skill shortage in cybersecurity.
- Used a multi-agent-based system with a hierarchical RAG architecture to solve the dynamic attack environment mapping and strategy identification in the pentesting processes.
- Used LLMs and RAG architectures along with techniques such as chain-of-thought reasoning, reinforcement learning, and in-context learning. Used OOP principles in coding and frameworks like PyTorch, LangChain, OpenAI, and Transformers.

Pen-Strategist : A Reasoning Framework for Penetration Testing [\[code\]](#)

2025 - 2026

- Proposed Pen-Strategist, a two-model framework for automated pentesting strategy derivation and action prediction, addressing limitations in strategy formulation, domain-specific reasoning, and tool selection in existing pentesting agents.
- Fine-tuned a Qwen-3-14B model using Group Relative Policy Optimization (GRPO) with a novel reasoning-centric dataset covering 240 HTB and VulnHub machines, achieving an 87% improvement in strategy derivation over the baseline and a 47.5% improvement in subtask completion in pentesting tasks.

ML based Network Traffic Analysis [\[code\]](#)

2023 - 2025

- Conducted research on encrypted network traffic analysis, focusing on the open-set classification problem where models must identify target traffic while rejecting unseen background traffic, addressing limitations of closed-set deep learning approaches in practical deployment scenarios.
- Developed TrafficLLM, a framework leveraging pre-trained LLMs (GPT-2, LLaMA-2-7B with QLoRA) as dataset-agnostic feature extractors for encrypted traffic traces, outperforming CNN-based and transformer-based baselines by up to 21.5% in open-set F1 score across seven datasets.

Honors Student's Project Supervision and Collaboration [\[code\]](#)

2025 - 2026

- Supervised and collaborated with honors students on SADE (Symptom-Aware Diagnostic Escalation), an LLM-based agentic framework for automated network fault diagnosis, encoding the classical Cisco troubleshooting methodology as an explicit multi-stage diagnostic policy.
- The framework follows a four-stage pipeline—initial reachability scan, deep layer-by-layer network scan (L2 through application layer), symptom to fault-family mapping, and skill-driven root cause localization—using a library of 15 Claude Skills and MCP tools to systematically isolate faults in Kathará-emulated network topologies.

RESEARCH INTERNSHIP

University of Sydney, Australia

Robust Open Set Classification for Encrypted Traffic Fingerprinting [\[code\]](#)

2021 - 2022

- Developed a novel network traffic classification method that can be deployed in resource constrained network devices using deep learning and statistical methods.
- The method could quantize classifier models to a 4-bit state with only 4% of performance degradation.

Quantifying and Exploiting Adversarial Vulnerabilities using Gradient-Based Input Pre-Filtering

2022

- Proposed a novel adversarial perturbation technique to optimize adversarial inputs for trained machine learning models using a novel technique called zero gradients.
- The method outperformed the state-of-the-art. We proved the transferability of our method between ML models using different datasets and shadow models.

HONORS RESEARCH

University of Moratuwa, Sri Lanka

Object Tracking, Reidentification and Activity Detection for Maritime Surveillance [\[code\]](#)

2022-2023

- Developed a thermal-based maritime surveillance system with real-time object tracking, suspicious activity detection, and viewpoint-invariant vessel re-identification, integrated into an interactive GUI for continuous 24×7 monitoring and alerting.
- The novel re-identification algorithm can re-identify vessels from different viewpoints, paying attention to even minor features of the vessel's shape. The work is under review.
- The activity detection algorithm identifies suspicious activities like human trafficking by processing the video feed in real time.

EXTERNAL RESEARCH

Monash University, Australia

Encoding and Decoding Data into DNA Using an Alphabet of DNA Sequences

2022

- Involved in a next generation technological research where we store data in high density DNA patterns.
- Researched on deep learning approaches to classify nanopore signals (small electric signals) corresponding to a given alphabet. Used CNN and LSTM architectures and obtained 95% accuracy.
- Used first order filters and Markov chains to optimize results for real world data.

- The project aims to identify anomalies of a multi-robot system using IMU, camera, and Lidar sensor data.
- Implemented an auto-encoder model to identify anomalies in IMU data of an autonomous robot.
- Used a GAN model to reconstruct the next frame of a frame sequence, thereby identifying anomalies in the camera sensor data.

OTHER PROJECTS

SMART BREADBOARD : TO FACILITATE REMOTE LAB EXPERIMENTS.

[Skills : Electronics, Arduino, IoT, Python]

Designed a smart breadboard using a MOSFET matrix to support virtual experiments during pandemic periods. Connections between columns and rows are made by switching on MOSFETs according to a user input. Users design circuits through the GUI, which are automatically implemented on the smart breadboard by switching MOSFET switches. The project won the IEEE CAS Student Design Competition 2020/21 in Sri Lanka and was selected for the IEEE Region 10 finals. [\[video\]](#)

ELDERBOT : ELDERLY CARE ELECTRONIC PRODUCT.

[Skills : Electronics, Arduino, HTML, Solidworks]

Designed an electronic product to assist elderly people in a fall or an emergency situation by sending alarm messages. Used an MPU6050 sensor, NodeMCU, and MQTT, and HTTP protocols. Developed software and a UI for user login and profile management. [\[video\]](#)

AUTONOMOUS MOBILE AND ASSISTING ROBOT FOR SLRC 2018. [\[CODE\]](#)

[Skills : Electronics, Arduino, Solidworks]

Designed and developed two robots. A mobile autonomous robot that follows lines and dashed lines, solves a line maze (using the DFS algorithm), collects, detects, sorts by color, and unloads coins. It communicates with an immobile assisting robot, which removes obstacles from its path.

IMAGE DOWNSAMPLING PROCESSOR DESIGN AND FPGA IMPLEMENTATION. [\[CODE\]](#)

[Skills : Verilog, FPGA]

Designed and implemented a RISC processor to downsample an image on the Altera DE2-115 FPGA board.

FILE COMPRESSION AND DECOMPRESSION ALGORITHM [\[CODE\]](#)

[Skills : C++]

Implement both code compression and decompression using C++ for a given input file. A customized rule set was used for the compression and decompression tasks.

PROFESSIONAL EXPERIENCE

Aug. 2024 Present	CASUAL ACADEMIC University of Sydney, Australia ➤ Conduct tutorials for the COMP5618, CSEC3616, CSEC5616, and INFO3616 Cybersecurity Engineering courses. ➤ Conduct tutorials for the INFO6007 Project Management in IT course.
Apr. 2024 Dec 2024	CASUAL ADMINISTRATIVE OFFICER - CYBER EVENT ORGANIZATION University of Sydney, Australia ➤ Conducted cybersecurity workshops for engineering students. Explained practical applications of theories.
Jul. 2023 Dec. 2023	SOFTWARE ENGINEER Axiata Digital Labs (Pvt) Ltd, Sri Lanka ➤ Worked on a project to build a low latency, high throughput API manager using C++ and Java. Used concepts such as OOP, lambda functions, api, multi-threaded coding, and used relevant tools.
Dec. 2021 Sep. 2022	STUDENT RESEARCH AFFILIATE University of Sydney, Australia ➤ Worked on two research projects on ML based network security. Studied and researched on traffic fingerprinting attacks, membership inference attacks, and model quantization effects. ➤ Completed and submitted two papers.
Jan. 2022 Jul. 2022	VISITING RESEARCHER Monash University, Australia ➤ Worked on a project based on developing a DNA alphabet to store data using the Nanopore technology. ➤ Used first-order filters for signal processing, statistical models like Markov models for classification. ➤ Developed machine learning models to identify DNA patterns of the alphabet with 96% accuracy with higher throughput.
Oct. 2022 Jul. 2023	VISITING INSTRUCTOR University of Moratuwa, Sri Lanka ➤ Conducted lab sessions for the “EN3143 : Electronic Control Systems” module. Explained theories and their applications.

HONORS AND AWARDS

Honorable Mention Award : International Physics Olympiad ‣ Organized by the International Physics Olympiad Board	2018
World Champions :IEEE ICAS 2021 Challenge (Montreal, Canada) : ‣ Organized by IEEE Signal Processing Society	2021
World Champions :IEEE IES Student Branch Chapter Competition : ‣ Organized by IEEE Industrial Electronics Society	2022
World Runner-up :IEEE International Humanitarian Technology Video Competition : ‣ Organized by IEEE Canada	2022
Represented Sri Lanka in the Asian Physics Olympiad ‣ Organized by the Physics Olympiad Committee of Vietnam	2018
NSW Connectivity Innovation Network(CIN) PhD scholarship ‣ Awarded by Connectivity Innovation Network, NSW, Australia	2023
Received funding from the USYD Collaborative Research Equipment Grant ‣ Received a grant for high-performance GPU cluster usage in the University of Sydney for 720 hours	2026
Champions : 2nd International Energy and Electricity Market Business Decision Competition ‣ Organized by the Shanghai University, China	2020
Nominee for the Most Outstanding Graduate of the Year 2023 ‣ University of Moratuwa, Sri Lanka	2023
Gold Medalist : Sri Lankan Physics Olympiad 2018 ‣ Organized by the Institute of Physics, Sri Lanka	2018
Dean's List in Semesters 1, 2, 4, 5, 6, 7, and 8 ‣ Faculty of Engineering, University of Moratuwa	2018 - 2023
Darrel Medal : The Most Outstanding Advanced Level Student Award ‣ Awarded by Richmond College, Galle, Sri Lanka	2017
Small Medal : Best Student in Mathematics ‣ Awarded by Richmond College, Galle, Sri Lanka	2015
School Colors for Cricket ‣ Awarded by Richmond College, Galle, Sri Lanka	2013

TECHNICAL SKILLS

PROGRAMMING LANGUAGES	Python, C/C++, Java, Verilog
LIBRARIES	Tensorflow, Pytorch, NumPy, Keras, Scikit-learn, Pandas, LangChain
SOFTWARE TOOLS	MATLAB, Altium, SOLIDWORKS, Spring Boot, Git, LaTeX, SonarQube, NodeRed
HARDWARE	Atmel AVR, Altera DE2, Raspberry Pi, NodeMCU

COURSES AND CERTIFICATION

Jun. 2020	IMPERIAL COLLEGE, LONDON, COURSERA ‣ Mathematics for Machine Learning-Linear Algebra
Aug. 2020	DEEPEARNING.AI ‣ Deep Learning Specialization – 5 Courses
Aug. 2023 - Present	EC COUNCIL ‣ Certified Ethical Hacker Certification

VOLUNTEERING AND LEADERSHIP SKILLS

IEEE Industrial Electronics Society - Student Branch Chapter of University of Moratuwa ‣ Chairman (2022-2023) and Vice Chairman (2021-2022) ‣ Provided leadership for a 150 member base to organize workshops, seminars, competitions, and social events. ‣ Emerged as “The best IES student branch chapter” in the worldwide competition in 2022. ‣ Won the “Emerging Student Branch Chapter” award in the IEEE Sri Lanka section awards – 2022.
Young Professionals Coordinator : Institute of Engineers Sri Lanka (IESL) NSW Chapter ‣ Main organizer HDR student gathering at Universities in Sydney. ‣ Organized a traditional event to celebrate the Sinhala and Hindu New Year.
HDR Student Representative : School of Computer Science, University of Sydney.

- Organized technical and non-technical HDR events to build the network and improve social skills in HDR students.

Mathematics Society, University of Moratuwa

- Chairman in 2020-2021
- Provided leadership for a 300 member base to organize seminars, workshops and competitions.

Co-chair of the Gold Chasers project - Rotaract Cub, University of Moratuwa

- Organized practical and tutorial sessions for the Sri Lankan Physics Olympiad team, 2019.
- The team obtained 2 bronze medals and one honorable mention in the International Physics Olympiad 2019.

Member of the Cricket Team, Richmond College Galle

- Won the all island championship in 2013 and semi-finalists in 2012.

REFERENCES

Dr. Suranga Seneviratne

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